Adaptive Image Denoising Approach for Low-Dose Computed Tomography

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***Abstract*—Low-dose computed tomography (LDCT) is usually performed by reducing the power of the x-ray tube in clinical CT scanners. However, images acquired through LDCT are known to be of low-quality due to the presence of statistical noise and other related artifacts. Effective denoising techniques are required to improve the quality of LDCT images towards green and safe CT imaging. In this paper, a new method is presented to improve the so-called, non-local means (NLM) filtering for effective LDCT imaging. The proposed method incorporates a prior knowledge obtained from probabilistic atlas during the filtering process. Additional anatomical information obtained through the atlas is likely to be useful in improving the image quality using NLM filtering. The proposed method is evaluated using real data and a notable improvement in image quality improvement is achieved.**

I. Introduction

Computed tomography (CT) imaging is employed through x-ray radiation of the target object such that the anatomical details and internal structures are visible. It is one of the most widespread non-invasive imaging modalities in medical diagnostics and interventional procedures. Nevertheless, high radiation dose resulted by being exposed to CT scanning may increase the risk of malignancy. The deleterious effects of the high radiation dose for CT scans become a major concern [1]. As guidelines of CT scanning, a low radiation should be as low as reasonably necessary. But in many cases, it is difficult to achieve good quality CT images. On the other hand, low-dose CT (LDCT) imaging may produce a noisy image and many artifacts, which lead to the deterioration of the diagnostic accuracy. So, there is a need to develop a technique which could denoise images in LDCT without losing image quality. Several techniques have been developed to deal with noise artifacts of LDCT imaging. Statistical iterative reconstruction (SIR) is known to be effective in removing noise artifacts. Moreover, high quality images would be achieved because of the ability to merge prior information to the image denoising method. The common prior knowledge used in this case can be classified into three categories as follows. (1) Information on the intensity value of the reconstructed image, (2) information regarding the pixel position and region boundaries, and (3) the one combining these two types.

Non-local means (NLM) filtering is an image processing technique developed for effective image denoising using information from the full image domain. Instead of using neighbor pixels to estimate the noise-free pixel values, it employs a technique to search for similar pixels with no locality restrictions. Several models are developed to use the NLM filtering in LDCT imaging and it is proved that it leads to superior image quality [2]–[6]. One major problem with the NLM filtering is that it considers the pixel intensity value as the only measure to validate the pixel without further consideration of spatial information.

In this paper, a new method to improve the NLM filtering algorithm in LDCT imaging is proposed. The proposed method considers prior knowledge resulting from probabilistic atlas that rely on only pixel position to further strengthen the NLM filtering. Evaluating by real data shows that the proposed method could enhance the image quality. The reason is that, the proposed method is based on combination of two different prior models to detect and reduce noise in LDCT. First, we consider the NLM as a prior information representing the intensity information. Second, the Probabilistic Atlas (PA) is used to define the missing pieces of data in image spatial domain. Therefore, both the intensity value and pixel position as considered in the LDCT image denoise problem. This paper is organized as follows. Section II is an overview of the original NLM algorithm, In section III, we describe the proposed method for LDCT image denoising. In section IV, we describe the experimental results. Finally, in section V, we discuss the results and present our conclusions.

II. Overview of NLM algorithm

The NLM filtering method was originally proposed by Buades *et al.* [7] as a non-iterative image denoising technique. In the non-local techniques, algorithms analyze data and collect the pixels from the whole image in order to look for similar characters. Assume that $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is an

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image, the NLM corresponding to image pixel x_i is a weighted sum of arbitrary image pixels x_j that satisfy some statistical properties.

$$NLM(x_i) = \sum_{j \in \mathbf{sw}} \omega_{ij} x_j, \tag{1}$$

where j is the position of a pixel located somewhere else in search window **sw** revolved around the pixel i. The weights satisfy the constrains $0 \leq \omega_{ij} \leq 1$ and $\sum_j \omega_{ij} = 1$ and defined as:

$$\omega_{ij} = \frac{1}{Z_i} \exp\left(-\frac{\|V_i - V_j\|_2^2}{h^2}\right), \tag{2}$$

where Z_i is a normalizing factor

$$Z_i = \sum_j \exp\left(-\frac{\|V_i - V_j\|_2^2}{h^2}\right), \tag{3}$$

V_i and V_j are two patches with the same size that centered at pixels x_i and x_j, h is a smoothing parameter and $\|.\|_2$ is the Euclidean distance between two patch windows.

III. Proposed method

The quality of images obtained from LDCT is low due to the effect of statistical noise. It is common to use NLM-based algorithms to reduce the image defect caused by the noise and other artifacts. The main drawback of the NLM-based technique is that it considers the pixel intensity as the only measure to define the anatomical structure similarities. This may incorrectly mix image pixels representing different anatomical structures. Probabilistic atlas, however, can provide a more accurate information of the pixel anatomy using data obtained from image dataset [8]. In this work, we develop non-local averaging technique using statistical models that represents additional prior information. The proposed framework is composed of two basic stages. First, we construct PA from images dataset obtained from various patients. Second, the PA is used as prior information together with the NLM for image denoising. The resulting adaptive technique, called Probabilistic-Atlas NLM (PA-NLM), is implemented by using PA information embedded in NLM filtering. The information obtained from the PA can be computed through two main stages, a learning phase followed by image denoising phase.

A. Stage I: Learning phase

In order to construct a probabilistic atlas first, we use an intensity-based automatic technique to co-register dataset images. The atlas is computed in the form of Gaussian mixture model (GMM) [9]. Second, the well-known EM algorithm [10], [11] is used to cluster the registered images into L components. A chart of learning stage is presented in Fig. 1. The density function of the GMM is expressed as:

$$p(x_i) = \sum_{l=1}^{L} \pi_l p(x_i|\Omega_l), i = 1, ..., n, \tag{4}$$

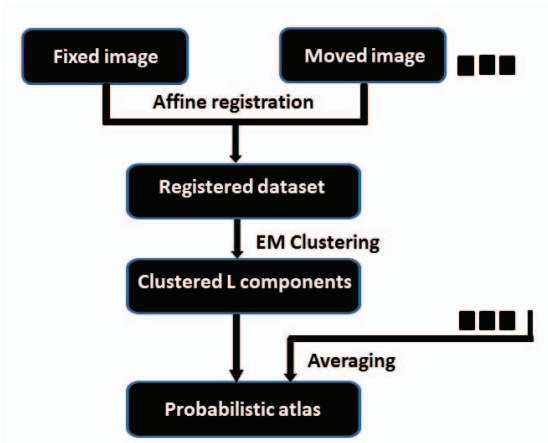


Fig. 1: Construction of the probabilistic atlas.

where n is the size of image vector, Ω_l $(l = 1, ..., L)$ is the class labels and π_l is the prior probability of the pixel x_i belonging to the mixture class Ω_l, which satisfies the following constraints.

$$0 \leq \pi_l \leq 1 \quad \text{and} \quad \sum_{l=1}^{L} \pi_l = 1. \tag{5}$$

The probability density function to each Gaussian distribution $p(x_i|\Omega_l)$ is expressed as:

$$p(x_i|\Omega_l) = \frac{1}{\sqrt{2\pi\sigma_l^2}} \exp\left(-\frac{(x_i - \mu_l)}{2\sigma_l^2}\right), \tag{6}$$

where $p(x_i|\Omega_l)$ is called a component of the mixture, μ_l and σ_l are mean value and covariance of mixture component l, respectively. The EM algorithm is used to evaluate and compute the GMM parameter (π_l, μ_l, σ_l).

B. Stage II: Image denoising phase

1) Atlas fitting: The PA should be aligned to fit with the image to increase atlas accuracy results. This process is implemented by clustering the noisy image into L components using the EM algorithm. Then, the PA is registered to the cluster version of the noisy image. The procedure used is shown in Fig. 2. The parameter ψ is the cluster index of noisy image and is defined as:

$$\psi_j = \{L : \max_{l=1}^{L} p(\Omega_l|x_j) \in L\}. \tag{7}$$

2) PA-NLM algorithm: The proposed denoising method is an adaptive combination of the NLM algorithm based on the average of image pixels that have similar statistical properties. The conventional NLM has been changed to be computed as follows:

$$NLM(x_i) = (1-\alpha)x_i + \alpha \sum_{x_j \in \mathbf{x}^{(t)}} \omega_{ij} x_j, \tag{8}$$

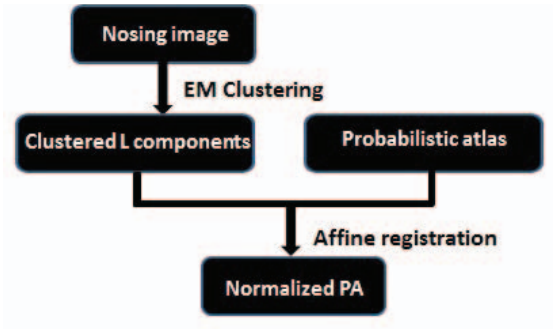


Fig. 2: A schematic of atlas fitting.

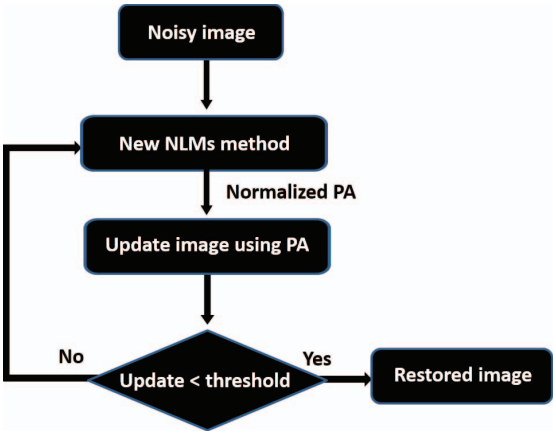


Fig. 3: A schematic of restored image using proposed method.

where $0 \leq \alpha \leq 1$ is a parameter controls the power of NLM regularization. The PA constructed from previous scans can be used to provide information regarding the pixel spatial information.

$$PA(x_i) = \frac{\sum_{(\gamma_j=\psi_j)\in L} \eta_j x_j}{\sum_{(\gamma_j=\psi_j)\in L} \eta_j}, \tag{9}$$

$$\eta_j = \max_{l=1}^{L} p(\Omega_l|x_j), \tag{10}$$

$$\gamma_j = \{L : \max_{l=1}^{L} p(\Omega_l|x_j) \in L\}, \tag{11}$$

where x_j define all pixels that belong to the same cluster L of the image pixel x_i. The proposed adaptive denoising method (Fig. 3) is performed through a combination of the two processes defined in Eqs. (8) and (11) as follows.

$$x_i^{(t+1)} = (1-\beta)NLM(x_i^{(t)}) + \beta PA(x_i^{(t)}), \tag{12}$$

where $0 \leq \beta \leq 1$ is the controling parameter to balance between the pixel intensity and pixel spatial information and t represents image update iteration number. It is clear that when β is relatively large (close to 1), the pixel position will have a large influence compared to the pixel intensity value and vice-verse.

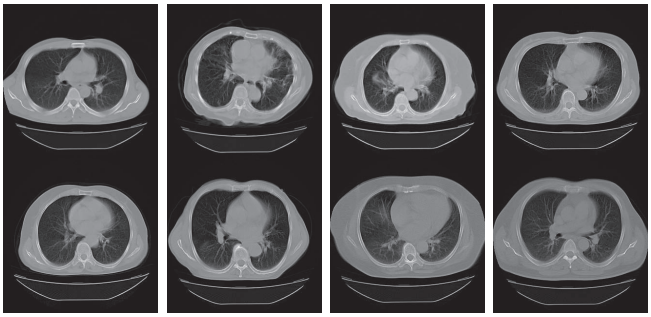
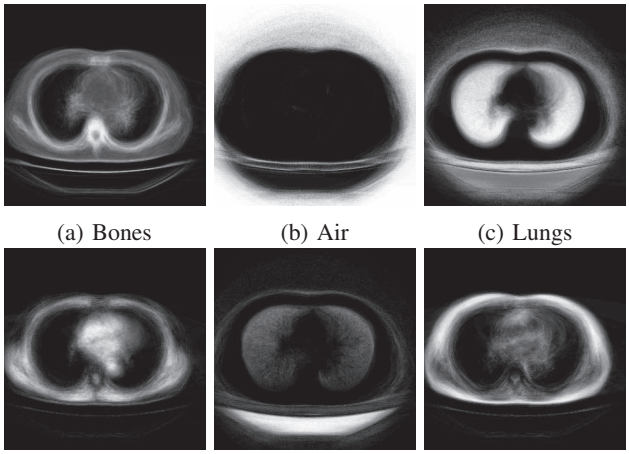


Fig. 4: An example demonstrates the large difference between dataset images.



(a) Bones (b) Air (c) Lungs

(d) Muscles (e) Bed (f) Fatty tissue

Fig. 5: Probabilistic atlas constructed from chest screening CT dataset.

IV. Experiment Results

A. Image quality measures

To estimate the image quality and evaluate the performance of the proposed method, we used the image quality measures of relative root mean square error (RRME).

$$RRME(\mathbf{x}) = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x_i^*)^2}{\sum_{i=1}^{n}(x_i^*)^2}} \tag{13}$$

where $\mathbf{x}^* = (x_1^*, x_2^*, \ldots, x_n^*)$ is the pixel value of true reference image and $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is the corresponding denoised image. Also, the peak signal-to-noise ratio (PSNR) is measured by

$$PSNR(\mathbf{x}) = 20\log_{10}\left(\frac{MAX(\mathbf{x})}{\sqrt{MSE(\mathbf{x})}}\right) \tag{14}$$

where MSE is the mean square error and is computed as follows:

$$MSE(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}(x_i - x_i^*)^2 \tag{15}$$

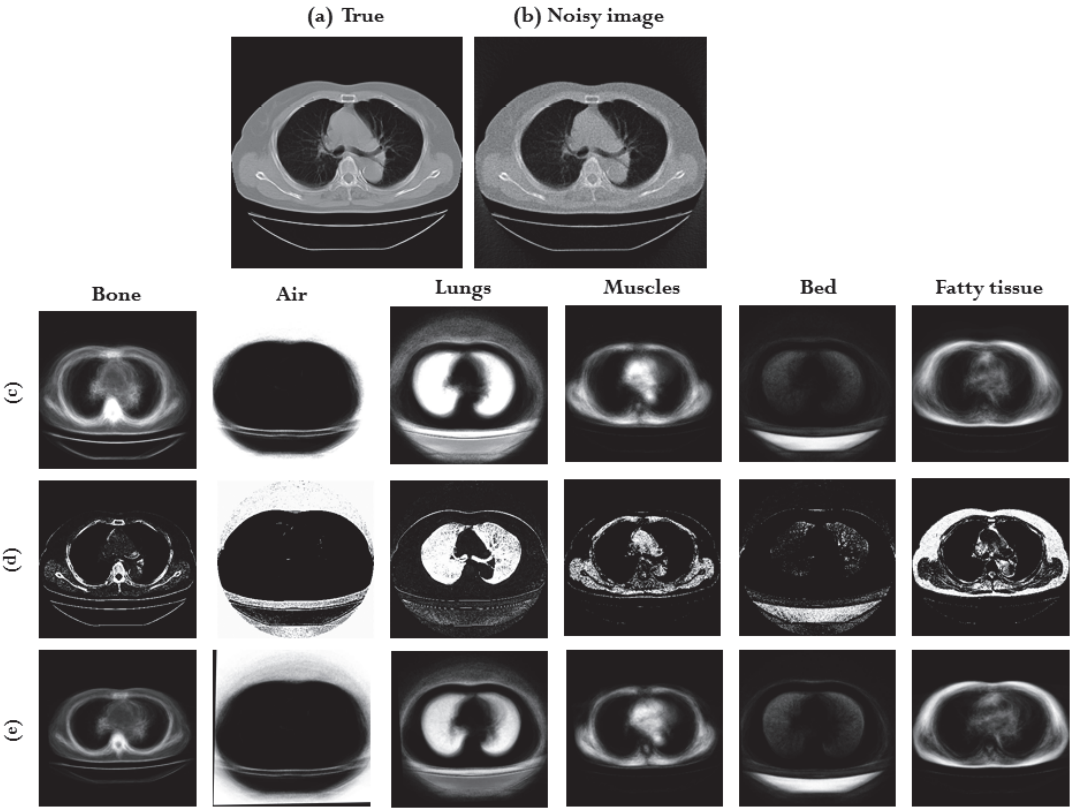


Fig. 6: (a) True image. (b) Noisy image. (c) Initial PA shown in Fig. 5 which matches inexactly with noisy image. (d) Clustered noisy image into L components. (e) PA after fitting initial PA with clustered noisy image for each of the L components.

B. Experiment setup

A dataset consists of 68 screening chest CT volumes of different patients is used to evaluate the proposed method[1]. Each volume consists of $25 - 31$ slices, where each slice composed of 320×320 pixels with pixel size of 1×1 mm. A randomly selected subset of 55 volumes were chosen to construct the PA. A sample of non-registreted images that demonstrate the large difference in anatomical information between the dataset images is shown in Fig. 4.

C. Atlas construction

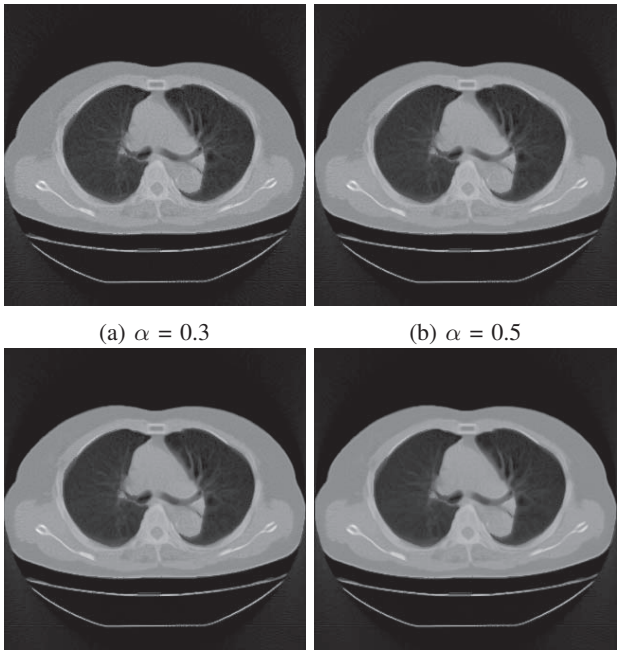
An arbitrary image is set as a reference image and all the remaining 54 images are registered to the reference image. The registration process is based on affine transformation with mean square error similarity metric. The EM algorithm is used to estimate the GMM parameters (π, μ, σ) and therefore the posterior probability function. The atlas is modeled to consist of six clusters ($L = 6$), which represent bones, air, lungs, muscles, bed and fatty-tissues. Figure 5 illustrates the atlas after 7 iterations of the EM algorithm. The initial noisy image obtained from a simulation of LDCT is computed. The PA shown in Fig. 5 is fitted with a clustered noisy image as shown in Fig. 6. The fitted atlas shown in Fig. 6(e) is used for the implementation of the PA-NLM algorithm.

[1]JAMIT medical image database, The Japanese Society of Medical Imaging Technology (JAMIT) (http://www.jamit.jp/cad/db/index.html).

D. Evaluation of PA-NLM parameters

The parameter α controls the power of NLM effect. In order to estimate the value of α in the proposed algorithm, we first fix $\beta = 0.3$ and the number of iterations $t = 2$ with which we can get a tentative high performance in noise removal. The value of α is changed from 0.3 to 0.9. A series of denoised images (Fig. 7) which confirm that when α be too small, it can not mitigate the noise. On the other hand, with a large value of α, noise are effectively eliminated. Moreover, evaluations using RRME and PSNR measurements are used to validate the effect of parameter α and results are illustrated in Fig. 8. It is also observed that the change of α has a small effect on the final image quality.

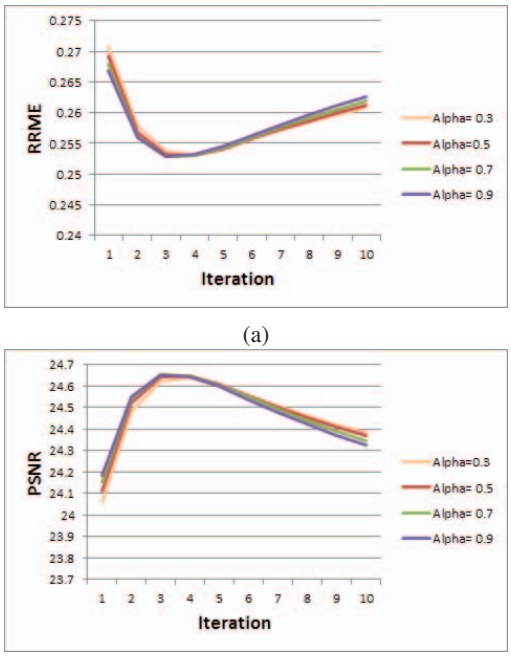
The parameter β controls the effect of PA. In order to estimate the effect of β in the proposed algorithm, we first set $\alpha = 0.7$ and fix the number of iterations $t = 2$. As shown in Fig. 9, increasing the value of β would suppress the noise which improve the image quality. However, this parameter should be carefully tuned to avoid over-smoothing artifacts that may be caused by high value of β. Also the RRME and PSNR measures are computed to evaluate the effect of parameter β as shown in Fig. 10.



(a) α = 0.3 (b) α = 0.5

(c) α = 0.7 (d) α = 0.9

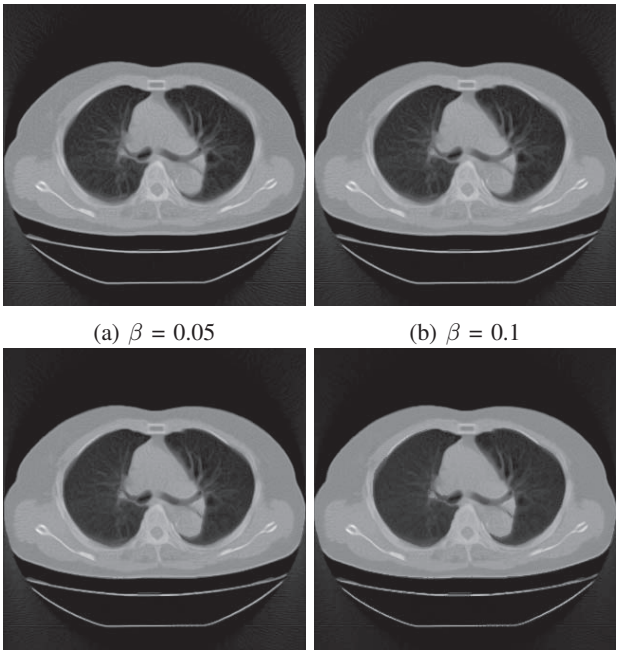
Fig. 7: Denoised images with $\beta = 0.3$ and $t = 2$ and different values of parameter α.



(a)

(b)

Fig. 8: RRME and PSNR with various α in case of $\beta = 0.3$ and $t = 2$. (a) RRME curves. (b) PSNR curves.



(a) β = 0.05 (b) β = 0.1

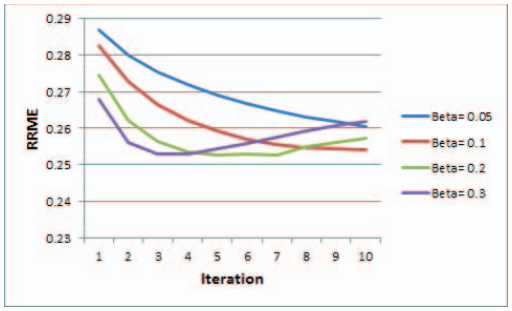
(c) β = 0.2 (d) β = 0.3

Fig. 9: Denoised images with various β in case of $\alpha = 0.7$ and $t = 2$.

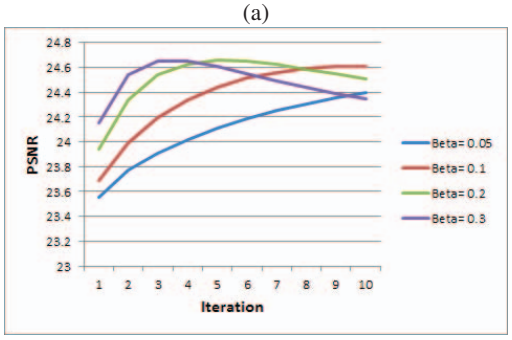
To evaluate how the number of iterations are contributing to the proposed algorithm, we conduct an experiment with different values of t. Figure 11 demonstrates the denoised results withvarious values of t for a given $\alpha = 0.5$ and $\beta = 0.07$. It shows that a small value of t (i.e. $t = 2$) has a relatively small effect to suppress the noise. With a large value of t ($t = 10$), we can observe an over-smoothing results especially in lung regions (see Fig. 11 (bottom row)). It is also observed that the denoised image has good quality in a small number of iterations ($t = 2$) on the contrary the effect of atlas is clearly showing in iterations ($t = 10$). This effect reflects the values assigned to α and β parameters. When these parameters are relatively small, it is needed to process a large number of iterations to get a satisfactory results. Taking computation time into account, it is better to set an appropriate large enough values of the parameters. After several attempts with different values of both α, β and t with various noisy structure images, it is found that the optimum values are $\alpha = 0.7$ $\beta = 0.3$ and $t = 2$. These results are shown in Fig. 12 and the image quality measurements are calculated and illustrated in Table 1.

TABLE I: Image quality measurements for denoised image shown in Fig. 12.

Method	RRME	PSNR
Noisy image	0.303484	23.0681
NLM	0.289268	23.4848
PA-NLM	0.256181	24.5398



(a)



(b)

Fig. 10: RRME and PSNR with various β in case of $\alpha = 0.7$ and $t = 2$. (a) RRME curves. (b) PSNR curves.

V. Conclusion

This paper presents a novel adaptive method for LDCT image denoise. The new method is based on a combinations of two different prior models to detect and reduce noise in LDCT. The NLM is considered as a prior information representing intensity information. Moreover, the PA is consider to define other missing pieces of data in image spatial domain. The proposed method has a strong power in detecting noise from the point of view that it handle prior information corresponding to both pixel value and pixel position. Experimental results demonstrate that the proposed method leads to a notable improvement in image quality.

References

[1] D. J. Brenner and E. J. Hall, "Computed tomography - an increasing source of radiation exposure," *New England Journal of Medicine*, vol. 357, no. 22, pp. 2277–2284, 2007.

[2] J. Ma, J. Huang, Q. Feng, H. Zhang, H. Lu, Z. Liang, and W. Chen, "Low-dose computed tomography image restoration using previous normal-dose scan," *Medical Physics*, vol. 38, no. 10, pp. 5713–5731, 2011.

[3] W. Xu and K. Mueller, "Efficient low-dose ct artifact mitigation using an artifact-matched prior scan," *Medical Physics*, vol. 39, no. 8, pp. 4748–4760, 2012.

[4] J. Ma, H. Zhang, Y. Gao, J. Huang, Z. Liang, Q. Feng, and W. Chen, "Iterative image reconstruction for cerebral perfusion ct using a pre-contrast scan induced edge-preserving prior," *Physics in Medicine and Biology*, vol. 57, no. 22, p. 7519, 2012.

[5] H. Zhang, J. Huang, J. Ma, Z. Bian, Q. Feng, H. Lu, Z. Liang, and W. Chen, "Iterative reconstruction for x-ray computed tomography using prior-image induced nonlocal regularization," *IEEE Transactions on Biomedical Engineering*, vol. 61, no. 9, pp. 2367–2378, 2014.

[6] H. Zhang, D. Zeng, H. Zhang, J. Wang, Z. Liang, and J. Ma, "Applications of nonlocal means algorithm in low-dose x-ray ct image processing and reconstruction: a review," *Medical Physics*, 2017.

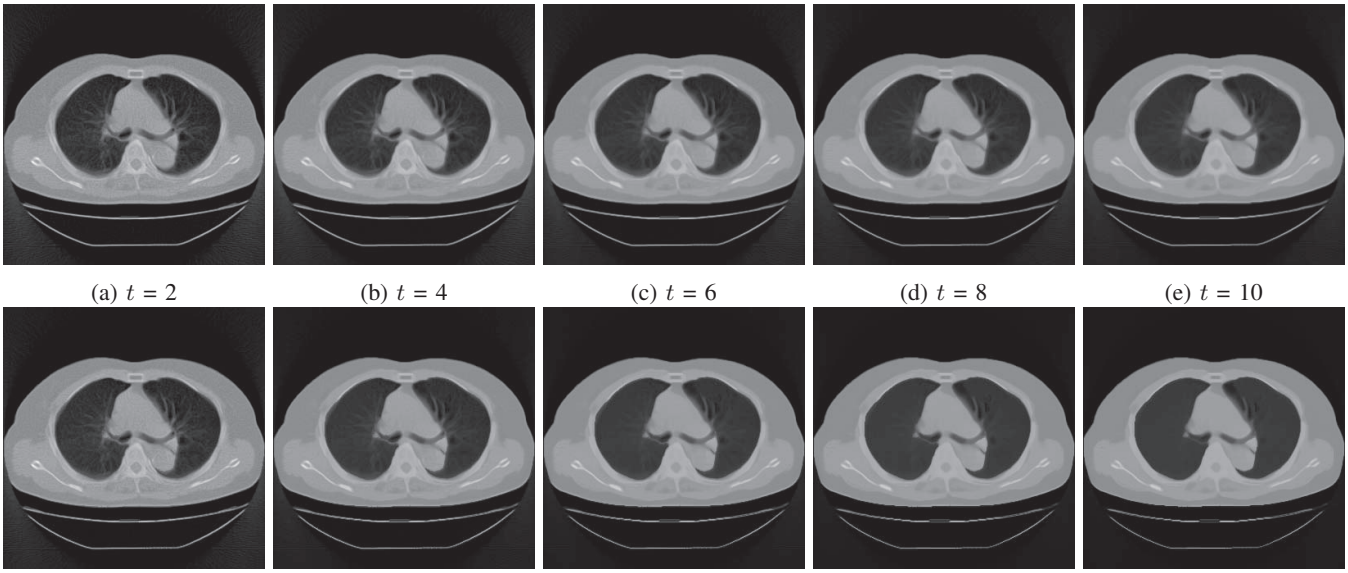
[7] A. Buades, B. Coll, and J.-M. Morel, "A review of image denoising algorithms, with a new one," *Multiscale Modeling & Simulation*, vol. 4, no. 2, pp. 490–530, 2005.

[8] E. A. Rashed and H. Kudo, "Probabilistic atlas prior for ct image reconstruction," *Computer Methods and Programs in Biomedicine*, vol. 128, pp. 119–136, 2016.

[9] D. M. Titterington, A. F. Smith, and U. E. Makov, *Statistical analysis of finite mixture distributions*. Wiley, 1985.

[10] G. McLachlan and T. Krishnan, *The EM algorithm and extensions*. John Wiley & Sons, 2007, vol. 382.

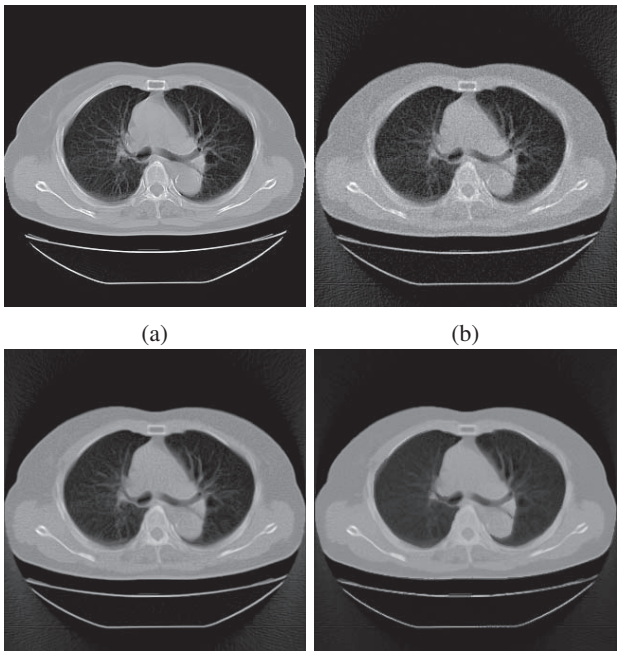
[11] M. A. T. Figueiredo and A. K. Jain, "Unsupervised learning of finite mixture models," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 3, pp. 381–396, 2002.



(a) t = 2 (b) t = 4 (c) t = 6 (d) t = 8 (e) t = 10

(f) t = 2 (g) t = 4 (h) t = 6 (i) t = 8 (j) t = 10

Fig. 11: Results of PA-NLM algorithm with different values of iteration number t: (a)-(e) are images obtained with $\alpha = 0.5$ and $\beta = 0.07$ and (f)-(j) are images obtained with $\alpha = 0.5$ and $\beta = 0.2$.



(a) (b)

(c) (d)

Fig. 12: (a) True image, (b) Noisy image, (c) Denoised image using standard NLM algorithm, and (d) Denoised image using proposed PA-NLM algorithm.